

SOME MIXTURES OF VON MISES DISTRIBUTIONS

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SUMMARY

Let $f(\cdot; \theta)$ be the density function of a random variable A , where the range of A is the points on the circumference of a circle. If the parameter θ is itself a random variable with density $g(\theta; \beta)$, where β is its parameter then the g -mixture of f is defined by $\int_{\theta} f \cdot g d\theta$. The Von Mises circular distribution is mixed with the sine-wave circular distribution, an asymmetrical distribution and with a Bessel function distribution and some apparently new circular distributions are obtained. The trigonometric moments of these distributions are derived.

1. Introduction

Let $T = \{F\}$ be a family of distributions and μ a measure on a Borel field of subsets of T with $\mu(T) = 1$. Then $H(\cdot) \doteq \int_T F(\cdot) d\mu(F)$ is again a distribution function called a μ mixture of T . In case of an absolutely continuous distribution the mixture can be written as $\int_{\theta} f(x, \theta) g(\theta) d\theta$, where $g(\theta)$ is the probability density function of the parameter θ which indexes the family $\{f\}$ of probability density functions.

It seems that very little has been done in the area of mixtures of circular distributions defined on the boundary of a circle. Recently, Gupta and Gupta (1974) has dealt with the problem of mixtures of sine-wave circular distribution.

The von Mises circular distribution has been the subject of active investigation by many. See Mardia (1972). In this paper we have obtained some mixtures of this distribution using a Bessel function distribution, sine-wave circular distribution and an asymmetric circular distribution as distributions of the parameters. To us, at least, two of these mixtures seem to be new circular distributions.

2. Some elements of the theory of circular distributions

Let Ω be the boundary of a unit circle. A circular distribution on Ω can be defined in the following way. Corresponding to a point on Ω , we can find a unique angle α , its vectorial angle restricted to a half-open interval I of length 2π and vice versa. Hence, the probability space Ω can be described as

$$(2.1) \quad \Omega = \{\alpha | \alpha \in (0, 2\pi]\}$$

We use the random angle A (capital alpha) and denote its distribution function by $F(\alpha)$. For an absolutely continuous distribution, which we confine to in this paper, we denote the density function of the random angle A by $f(\alpha)$ which is the derivative of $F(\alpha)$.

2.1 Trigonometric Moments

Definition. The trigonometric moment of a circular distribution of the random angle A having the density function $f(\alpha)$ are the functions defined, for integer values ν , by

$$(2.2) \quad \Phi_f(\nu) = \int_0^{2\pi} \exp(i\nu\alpha) f(\alpha) d\alpha.$$

It follows that $\Phi_f(0) = 1$ and $\Phi_f(-\nu)$ equals the complex conjugate $\overline{\Phi_f(\nu)}$ of $\Phi_f(\nu)$. Also $|\Phi_f(\nu)| \leq 1$ for all ν . For more properties of $\Phi_f(\nu)$, see Batschelet (1963) and Shohat and Tamarkin (1963). Hurwitz (1903) showed that the trigonometric moments determine a circular distribution uniquely.

2.2 Some Distributions

2.2.1 Von Mises Circular Distribution

For statistical applications von Mises (1918) proposed a circular distribution having the density function

$$(2.3) \quad f(\alpha) = \frac{\exp[k \cos(\alpha - \alpha_0)]}{2\pi I_0(k)}, \quad \alpha, \alpha_0 \in (0, 2\pi], k \geq 0,$$

where $I_\nu(k)$, for any integer ν , denotes the ν -th Bessel function with purely imaginary argument, and k and α_0 are the parameters. For this distribution the trigonometric moments are given by

$$\Phi_f(0) = 1 \text{ and } \Phi_f(\nu) = [I_\nu(k)/I_0(k)] \exp(i\nu\alpha_0), \nu \neq 0.$$

2.2.2 Sine-wave Circular Distributions

This circular distribution has the density function

$$(2.4) \quad f(\alpha) = \left(\frac{1}{\pi}\right) [\frac{1}{2} + \rho \cos(\alpha - \theta)], \quad \alpha, \theta \in (0, 2\pi], \quad 0 \leq \rho \leq \frac{1}{2}.$$

Its trigonometric moments are given by

$$\Phi_f(0) = 1, \quad \Phi_f(\nu) = \rho \exp(-i\nu\theta), \quad \nu = \pm 1$$

and

$$\Phi_f(\nu) = 0, \nu = \pm 2, \pm 3, \dots$$

The non-zero moments $\Phi_f(0)$, $\Phi_f(1)$ and $\Phi_f(-1)$ are about $\alpha = \theta$, $\theta \neq 2\pi$. A linear representation of this density function is a full period of a sine curve and hence the name.

2.2.3 An Asymmetric Distribution

As an example of an asymmetric circular distribution we shall use the density function

$$(2.5) \quad f(\alpha) = (\frac{1}{2}\pi) [1 + \rho(\alpha - \pi)], \alpha \in (0, 2\pi], 0 < \rho \leq 1/\pi.$$

For this distribution the trigonometric moments are given by $\Phi_f(0) = 1$ and $\Phi_f(\nu) = (-1)^{\nu-1} i\rho/\nu, \nu \neq 0$.

2.2.4 A Bessel Function Distribution

We shall use the distribution with the density function

$$(2.6) \quad g(k) = \exp[-\frac{1}{2}(1 + k^2)] k I_0(k), k > 0,$$

where $I_0(k)$ is as defined in 2.2.1.

3. Mixtures of von Mises circular distribution

Let $f(\alpha, \theta)$, θ being a parameter, be the density function of a random variable A . Suppose the parameter θ is itself a random variable with the density function $g(\theta, \beta)$, β being a parameter, called 'Mixing' density function. Then the density function $h(\alpha, \beta)$ given as

$$(3.1) \quad h(\alpha, \beta) = \int_{\theta} f(\alpha, \theta) g(\theta, \beta) d\theta$$

is called the 'g-mixture' of the density function $f(\alpha, \theta)$.

We now state a lemma which establishes a relationship between the trigonometric moments of the original density function $f(\alpha, \theta)$ and the g-mixture $h(\alpha, \beta)$. This lemma will be used here to obtain the trigonometric moments of the mixtures.

Lemma 1:

Let $\Phi_f(\nu, \theta)$ and $\Phi_h(\nu)$ denote the ν -th trigonometric moments, respectively, of the density functions $f(\alpha, \theta)$ and $h(\alpha, \beta)$. Then

$$(3.2) \quad \Phi_h(\nu) = \int_{\theta} \Phi_f(\nu, \theta) g(\theta, \beta) d\theta.$$

We now give three mixtures of the von Mises distribution.

3.1 Mixing with sine-wave circular distribution

We write the density function of von Mises distribution as

$$(3.3) \quad f(\alpha, \alpha_0) = \frac{\exp[k \cos(\alpha - \alpha_0)]}{2\pi I_0(k)}, \quad \alpha, \alpha_0 \in (0, 2\pi], k > 0,$$

where k is fixed and α_0 has the sine-wave distribution having the 'mixing' density function

$$(3.4) \quad g(\alpha_0, \rho, \beta) = (1/\pi)(\frac{1}{2} + \rho \cos(\alpha_0 - \beta)), \quad \beta, \alpha_0 \in (0, 2\pi], 0 \leq \rho \leq \frac{1}{2},$$

ρ and β being known parameters. The g-mixture of the density function $f(\alpha, \alpha_0)$ is

$$\begin{aligned} (3.5) \quad h(\alpha, \rho, \beta) &= \int_0^{2\pi} \frac{\exp(k \cos(\alpha - \alpha_0))}{2\pi I_0(k)} \cdot (1/\pi)(\frac{1}{2} + \rho \cos(\alpha_0 - \beta)) d\alpha_0 \\ &= (1/2\pi) \int_0^{2\pi} \frac{\exp(k \cos(\alpha - \alpha_0))}{2\pi I_0(k)} d\alpha_0 \\ &\quad + (\rho/\pi) \operatorname{Re} \left[\exp(-i\beta) \int_0^{2\pi} \frac{\exp(i\alpha_0 + k \cos(\alpha - \alpha_0))}{2\pi I_0(k)} d\alpha_0 \right]. \end{aligned}$$

The first integral on the right of the relation (3.5) equals 1, since the integrand can be treated as a von Mises density function of A_0 for fixed α . The second integral equals the first trigonometric moment $\Phi_1(1)$ of the random angle A_0 having the von Mises distribution with fixed α . Hence from section (2.2.1), (with the roles of α and α_0 interchanged), we have

$$\begin{aligned} (3.6) \quad h(\alpha, \rho, \beta) &= (1/2\pi) + (\rho/\pi) \operatorname{Re} \left[\frac{I_1(k)}{I_0(k)} \exp(i(\alpha - \beta)) \right] \\ &= (1/\pi) \left[\frac{1}{2} + \rho \frac{I_1(k)}{I_0(k)} \cos(\alpha - \beta) \right]. \end{aligned}$$

This shows that the mixture is a sine-wave distribution with parameters

$$\beta \text{ and } \rho \frac{I_1(k)}{I_0(k)}, \quad 0 \leq \rho \frac{I_1(k)}{I_0(k)} \leq \frac{1}{2}, \quad \beta \in (0, 2\pi].$$

Lemma 1 and the property of uniqueness of trigonometric moments also enables us to reach the above conclusion.

3.2 Mixing with an asymmetric distribution

As before, we write the density function of von Mises distribution as

$$f(\alpha, \alpha_0) = \frac{\exp(k \cos(\alpha - \alpha_0))}{2\pi I_0(k)}, \alpha, \alpha_0 \in (0, 2\pi], k > 0,$$

where k is fixed and α_0 has the density function

$$(3.7) \quad g(\alpha_0, \rho) = (1/2\pi)(1 + \rho(\alpha_0 - \pi)), \alpha_0 \in (0, 2\pi], 0 < \rho \leq 1/\pi,$$

ρ being a fixed known parameter. The g-mixture of the density function $f(\alpha, \alpha_0)$ is

$$(3.8) \quad h(\alpha, \rho) = \int_0^{2\pi} \frac{\exp(k \cos(\alpha - \alpha_0))}{2\pi I_0(k)} \cdot (1/2\pi)(1 + \rho(\alpha_0 - \pi)) d\alpha_0$$

$$= (1/2\pi) - \rho/2 + (\rho/2\pi) \int_0^{2\pi} \frac{\exp(k \cos(\alpha - \alpha_0))}{2\pi I_0(k)} \alpha_0 d\alpha_0.$$

The last integral, by using the transformation

$$\gamma = \begin{cases} \alpha_0 - \alpha, & \gamma \in (0, 2\pi - \alpha] \\ \alpha_0 - \alpha + 2\pi, & \gamma \in (2\pi - \alpha, 2\pi] \end{cases}$$

equals to $\alpha + \pi - 2\pi(1 - F(2\pi - \alpha))$, where $F(\cdot)$ is the cumulative distribution function of von Mises distribution. (This has been tabulated for various values of k . See Batschelet (1965)). Hence

$$(3.9) \quad h(\alpha, \rho) = (1/2\pi)(1 + \rho(\alpha - 2\pi(1 - F(2\pi - \alpha))))), \alpha \in (0, 2\pi].$$

This seems to be a new circular distribution.

By using lemma 1, the trigonometric moments of the g-mixture are obtained as

$$(3.10) \quad \Phi_h(\nu) = \int_0^{2\pi} \frac{I_\nu(k)}{I_0(k)} \exp(i\nu\alpha_0) \cdot (1/2\pi)(1 + \rho(\alpha_0 - \pi)) d\alpha_0, \nu \neq 0$$

$$= -\frac{I_\nu(k)}{I_0(k)} \cdot \frac{\rho}{\nu} i, \nu \neq 0$$

and $\Phi_h(0) = 1$.

3.3 Mixing with a Bessel Function distribution

Here we treat $k > 0$ as a random variable with the density function in (2.6) and keep α_0 fixed in the density function of von Mises distribution given in (2.3). Then

$$(3.11) \quad f(\alpha, k) = \frac{\exp(k \cos(\alpha - \alpha_0))}{2\pi I_0(k)}, \quad k > 0, \alpha, \alpha_0 \in (0, 2\pi]$$

and

$$(3.12) \quad g(k) = \exp(-\frac{1}{2}(1 + k^2))k I_0(k), \quad k > 0.$$

The g -mixture of the density function $f(\alpha, k)$ is

$$\begin{aligned} (3.13) \quad h(\alpha) &= \int_0^\infty \frac{\exp(k \cos(\alpha - \alpha_0))}{2\pi I_0(k)} \cdot \exp(-\frac{1}{2}(1 + k^2))k I_0(k) dk \\ &= (1/2\pi) \int_0^\infty k \exp(-\frac{1}{2}(k^2 - 2k \cos(\alpha - \alpha_0) + 1)) dk \\ &= (1/2\pi) \exp(-\frac{1}{2} \sin^2(\alpha - \alpha_0)) \int_0^\infty k \exp(-\frac{1}{2}(k - \cos(\alpha - \alpha_0))^2) dk \end{aligned}$$

To evaluate the last integral, we put $k - \cos(\alpha - \alpha_0) = u$ obtaining its value as

$$\exp(-\frac{1}{2} \cos^2(\alpha - \alpha_0)) + \cos(\alpha - \alpha_0) \cdot \sqrt{2\pi} (1 - \Phi(-\cos(\alpha - \alpha_0))),$$

where $\Phi(\cdot)$ is the usual cumulative normal distribution function. Hence the g -mixture is

$$(3.14) \quad h(\alpha) = (1/2\pi) \exp(-\frac{1}{2}) + \frac{\cos(\alpha - \alpha_0)}{\sqrt{2\pi}} \exp(-\frac{1}{2} \sin^2(\alpha - \alpha_0)) (1 - \Phi(-\cos(\alpha - \alpha_0))),$$

$\alpha, \alpha_0 \in (0, 2\pi].$

The above probability density function also seems to be a new circular distribution.

The easiest method to obtain its trigonometric moments is to use the lemma 1.

Here

$$\begin{aligned} \Phi_h(\nu) &= \int_0^\infty \frac{I_\nu(k)}{I_0(k)} \exp(i\nu\alpha_0) \exp(-\frac{1}{2}(1 + k^2))k I_0(k) dk \\ &= \exp(i\nu\alpha_0 - \frac{1}{2}) i^{-\nu} \int_0^\infty k \exp(-k^2/2) J_\nu(ik) dk, \end{aligned}$$

where $J_\nu(\cdot)$ is Bessel function of order ν ,

$$= \exp(i\nu\alpha_0 - \frac{1}{2}) i^{-\nu} \frac{\Gamma(\nu/2 + 1)(i/\sqrt{2})^\nu}{\Gamma(\nu + 1)} \cdot F_1(\nu/2 + 1; \nu + 1; \frac{1}{2}),$$

by using formula 2 of section 13.3 on p. 393 of Watson (1962), or

$$(3.15) \quad \Phi_h(\nu) = \frac{\exp(i\nu\alpha_0 - \frac{1}{2})}{(\sqrt{2})^\nu} \cdot \frac{\Gamma(\nu/2 + 1)}{\Gamma(\nu + 1)} \cdot F_1(\nu/2 + 1; \nu + 1; \frac{1}{2}).$$

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